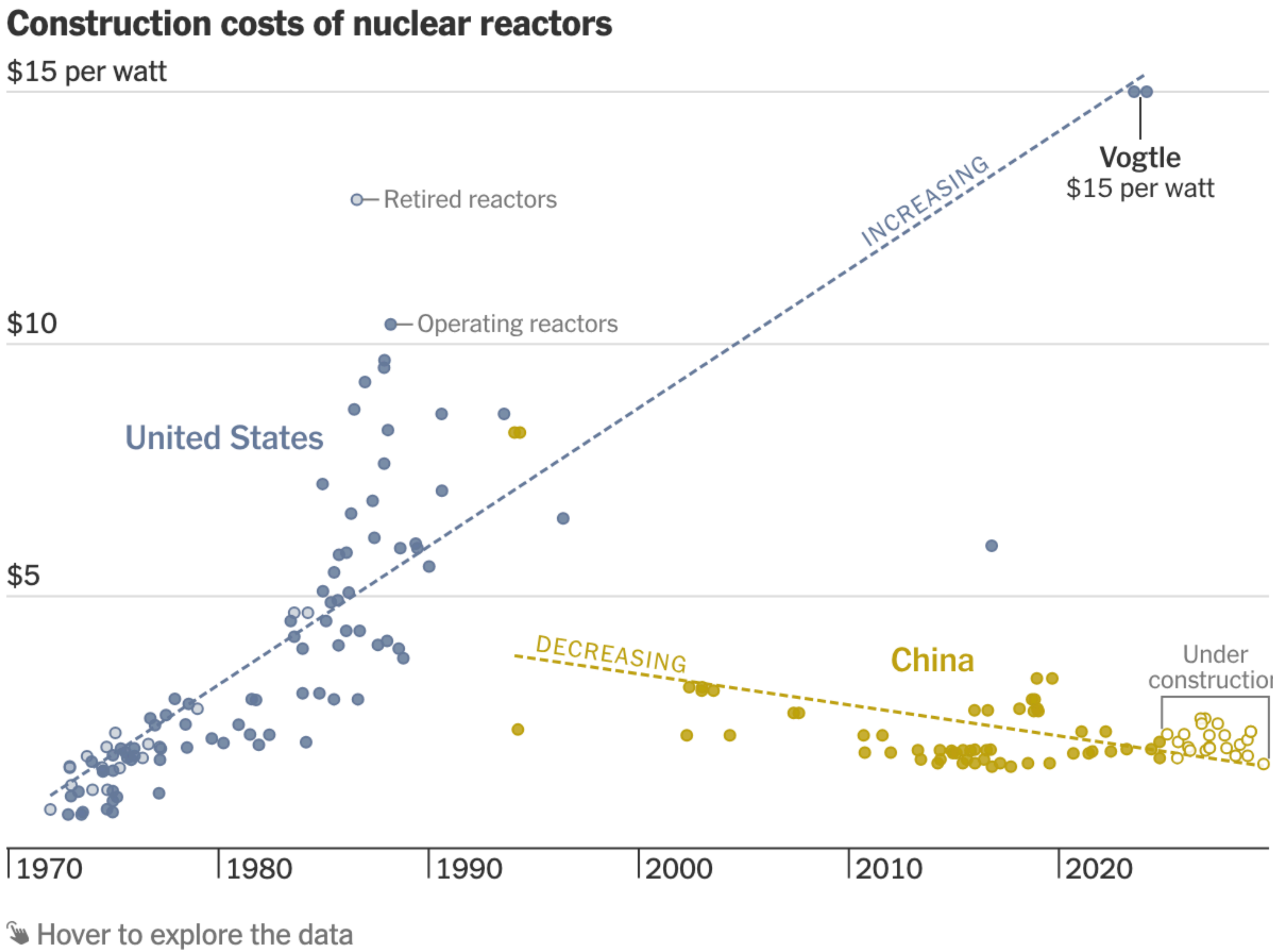
Assumptions, diagnostics, and multicollinearity

Activity



Construction costs of nuclear reactors
$15 per watt
$10
$5
Retired reactors
Operating reactors
United States
INCREASING
Vogtle
$15 per watt
DECREASING
China
Under
construction
1970
1980
1990
2000
2010
2020
Hover to explore the data

Data

Data on the number of doctors, hospitals, and beds in 53 randomly selected US counties. Variables include:

- `MDs`: number of doctors in the county
- `Hospitals`: number of hospitals in the county
- `Beds`: number of beds in the hospitals

Goal: Predict the number of doctors using the number of hospitals and the number of beds

Multiple regression model

$$\text{MDs} = \beta_0 + \beta_1 \text{ Hospitals} + \beta_2 \text{ Beds} + \varepsilon$$

What assumptions does our multiple regression model make?

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> What assumptions does our multiple regression model make?

The same assumptions as before:

- Shape
- Constant variance
- Normality
- Independence
- Randomness

Shape and constant variance

$$\text{MDs} = \beta_0 + \beta_1 \text{ Hospitals} + \beta_2 \text{ Beds} + \varepsilon$$

Shape: the shape of the regression model is at least approximately correct

- In this case, the number of doctors has a linear relationship with the number of hospitals and the number of beds

Constant variance: The variance of the noise term ε is the same for all numbers of hospitals and beds

> How do we assess the shape and constant variance assumptions?

Shape and constant variance

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A residual plot!

Residual plot

$$\widehat{\text{MDs}} = -472.08 + 117.40\ \text{Hospitals} + 1.26\ \text{Beds}$$

> What goes on the x and y axes of my residual plot?

Residual plot

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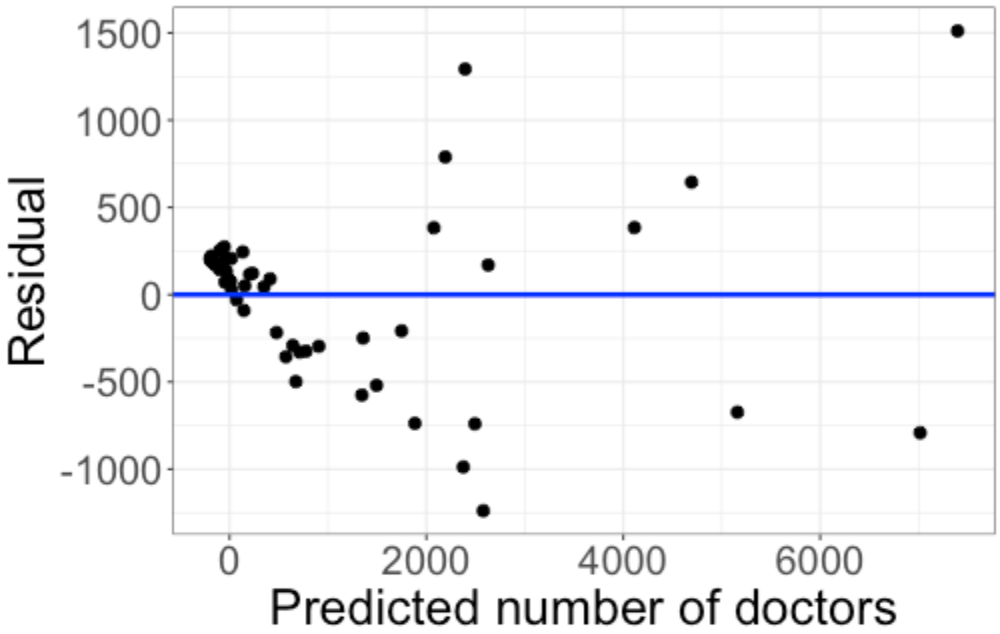
> What goes on the x and y axes of my residual plot?

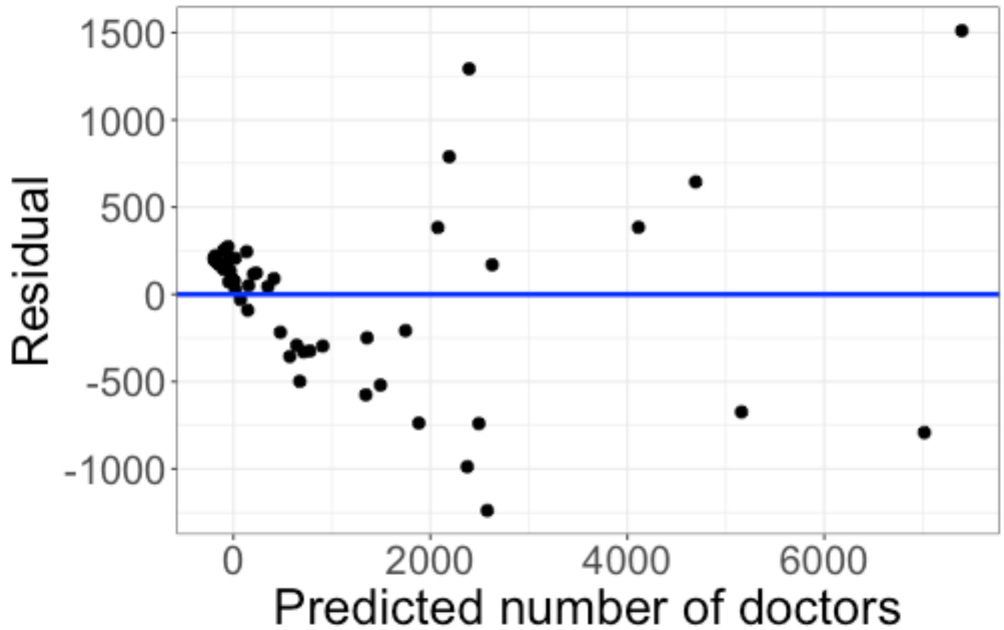
- x axis: the predicted response ($\widehat{\text{MDs}}$)
- y axis: the residuals ($\text{MDs} - \widehat{\text{MDs}}$)

Residual plot



How do the shape and constant variance assumptions look?

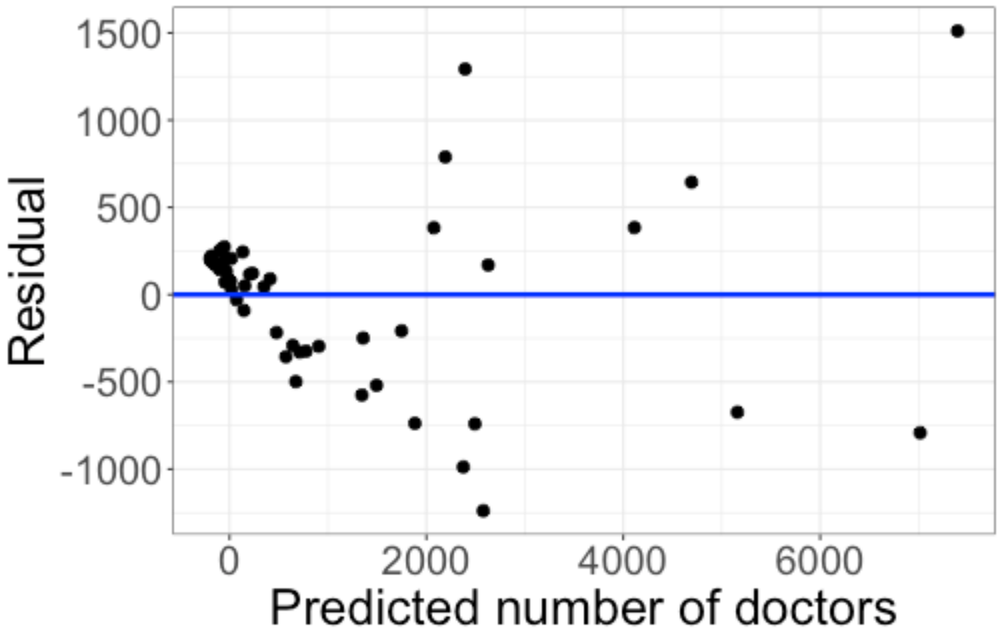
Residual plot



> How do the shape and constant variance assumptions look?

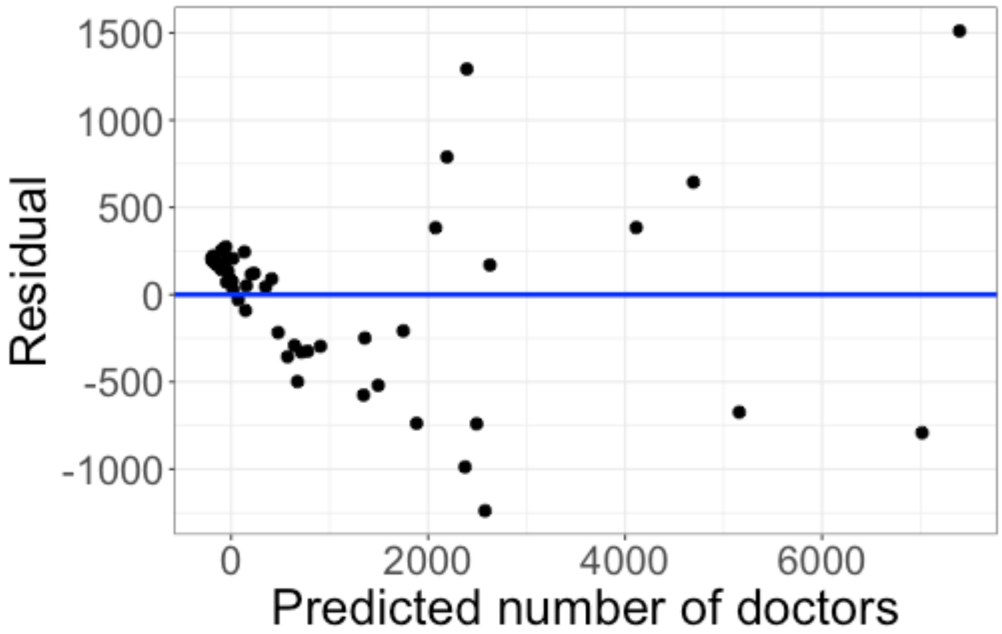
Not good -- violations to both shape and constant variance

Residual plot



> How do we address violations of shape and constant variance?

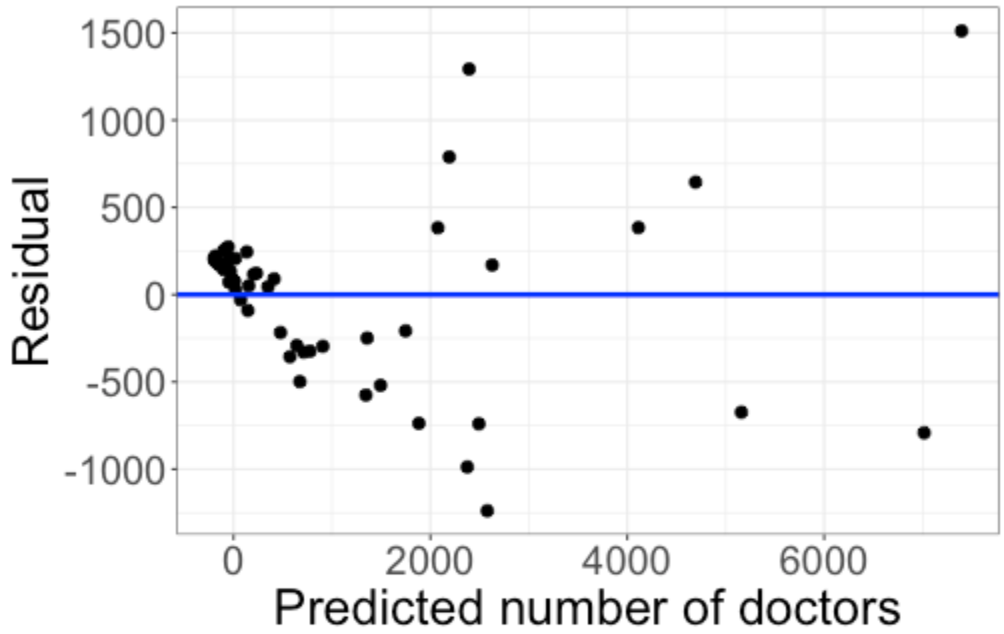
Residual plot



> How do we address violations of shape and constant variance?

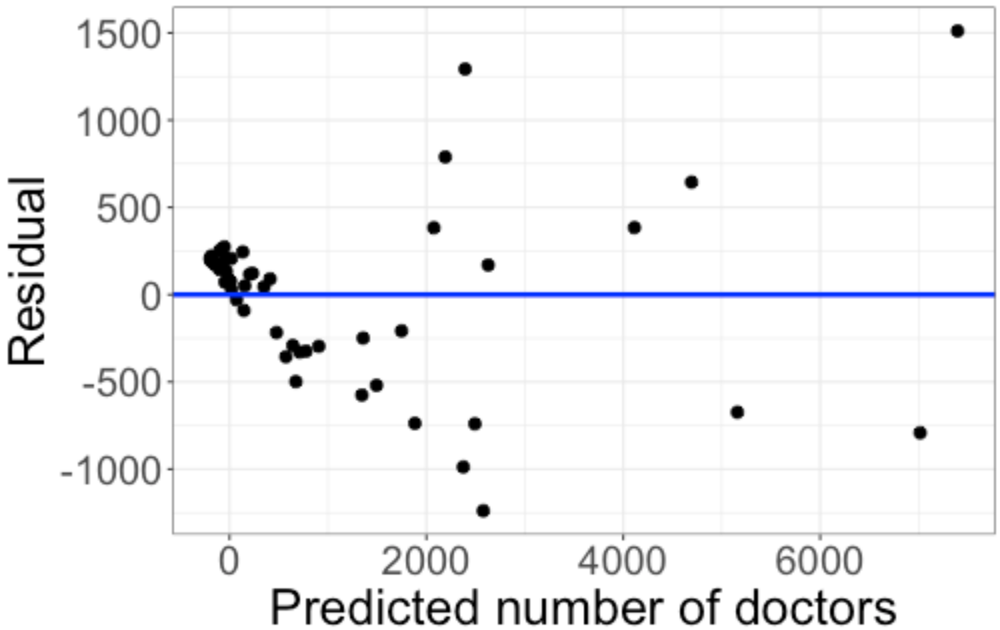
Transformations!

Residual plot



What transformation would you recommend?

Residual plot

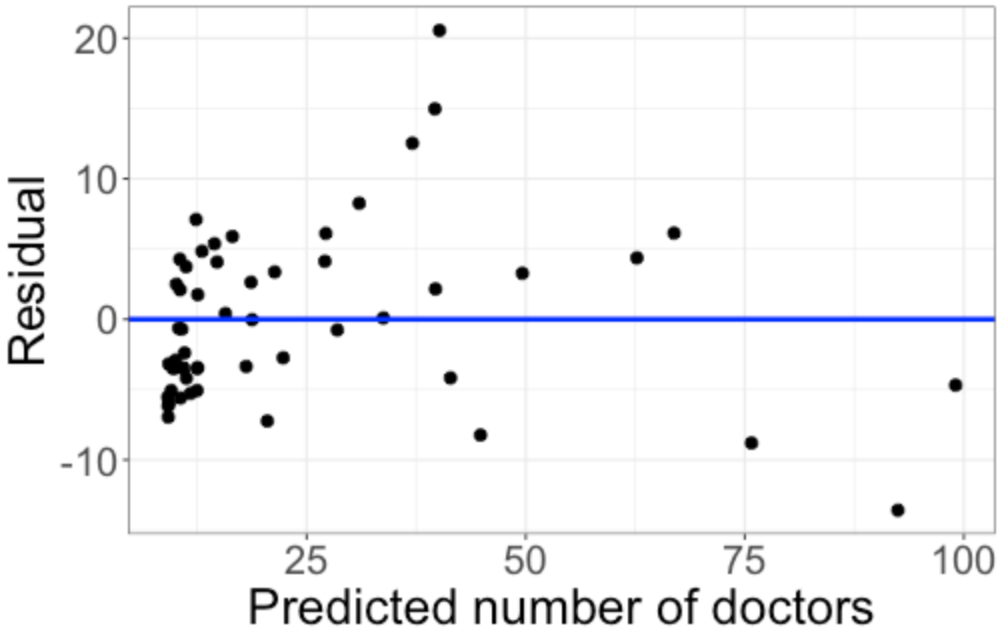


> What transformation would you recommend?

Maybe log or square root the response

Transforming the data

$$\widehat{\sqrt{\text{MDs}}} = 3.58 + 2.58 \text{ Hospitals} + 0.012 \text{ Beds}$$



This looks better!

Normality

$$\sqrt{\text{MDs}} = \beta_0 + \beta_1 \text{ Hospitals} + \beta_2 \text{ Beds} + \varepsilon$$

Normality assumption: the noise term ε follows a normal distribution

> How do we assess the normality assumption?

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> How do we assess the normality assumption?

With a QQ plot!

QQ plot

$$\widehat{\sqrt{\text{MDs}}} = 3.58 + 2.58\ \text{Hospitals} + 0.012\ \text{Beds}$$

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QQ plot

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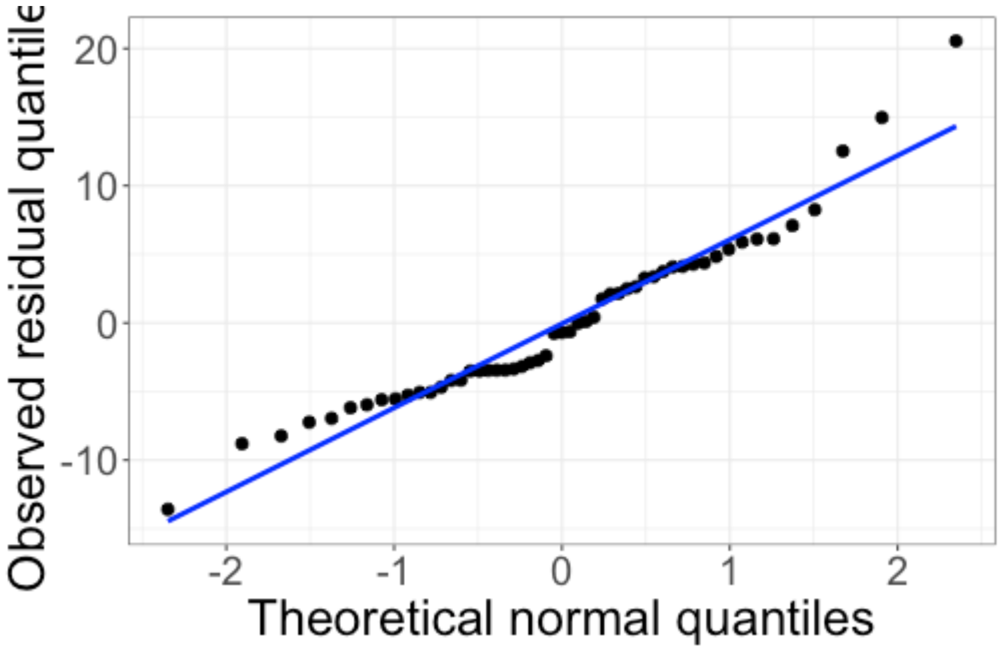
> What goes on the x and y axes of my QQ plot?

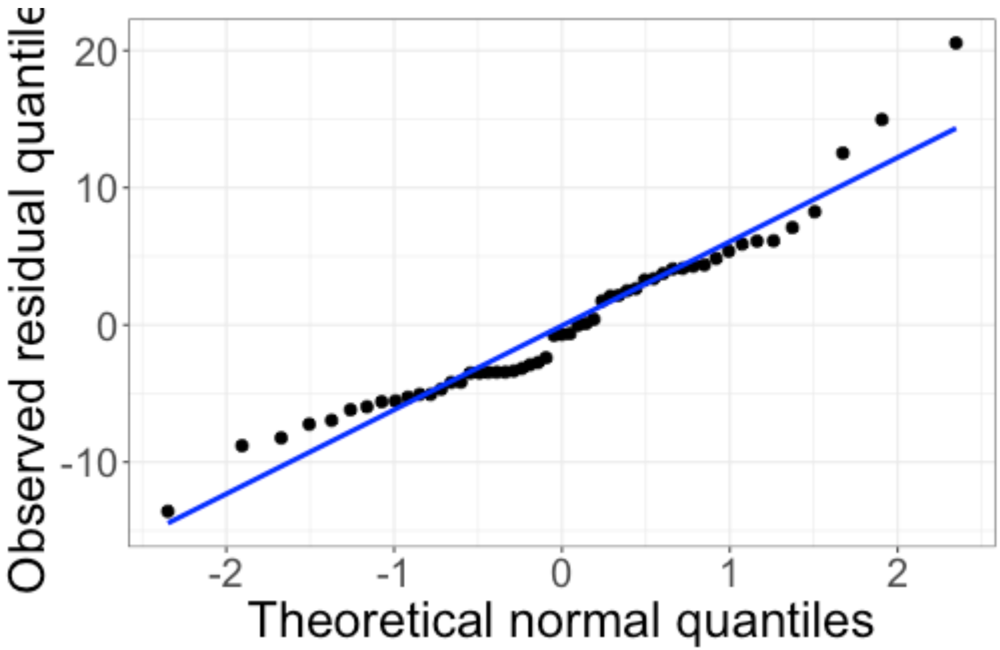
- x axis: theoretical normal quantiles
- y axis: observed residual quantiles

QQ plot



How does the QQ plot look?

QQ plot



How does the QQ plot look?

Pretty good!

Independence and randomness

$$\sqrt{\text{MDs}} = \beta_0 + \beta_1 \text{ Hospitals} + \beta_2 \text{ Beds} + \varepsilon$$

Independence assumption: Observations are independent -- one point falling above or below the line has no influence on the location of another point

Randomness assumption: The data are obtained using a random process, such as a random sample or randomized experiment

> How do we assess the independence and randomness assumptions?

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> How do we assess the independence and randomness assumptions?

Think about the data generating process (can't check these assumptions with plots)

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> Do independence and randomness seem reasonable for the county health data?

Independence and randomness

Independence assumption: Observations are independent -- one point falling above or below the line has no influence on the location of another point

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> Do independence and randomness seem reasonable for the county health data?

- Independence: reasonable to assume the number of hospitals/doctors for one county doesn't affect the number of hospitals/doctors for another county
- Randomness: we are told the data come from a random sample

Comparing models

With both variables:

- $\widehat{\sqrt{\text{MDs}}} = 3.58 + 2.58 \text{ Hospitals} + 0.012 \text{ Beds}$
- $R^2_{adj} = 0.919$

With just Beds:

- $\widehat{\sqrt{\text{MDs}}} = 8.49 + 0.018 \text{ Beds}$
- $R^2_{adj} = 0.899$

With just Hospitals:

- $\widehat{\sqrt{\text{MDs}}} = -2.75 + 6.88 \text{ Hospitals}$
- $R^2_{adj} = 0.849$

Comparing models

> Each variable is strongly associated with the response, but using both in the model doesn't improve the R^2_{adj} by much.
>
> Why the small change?

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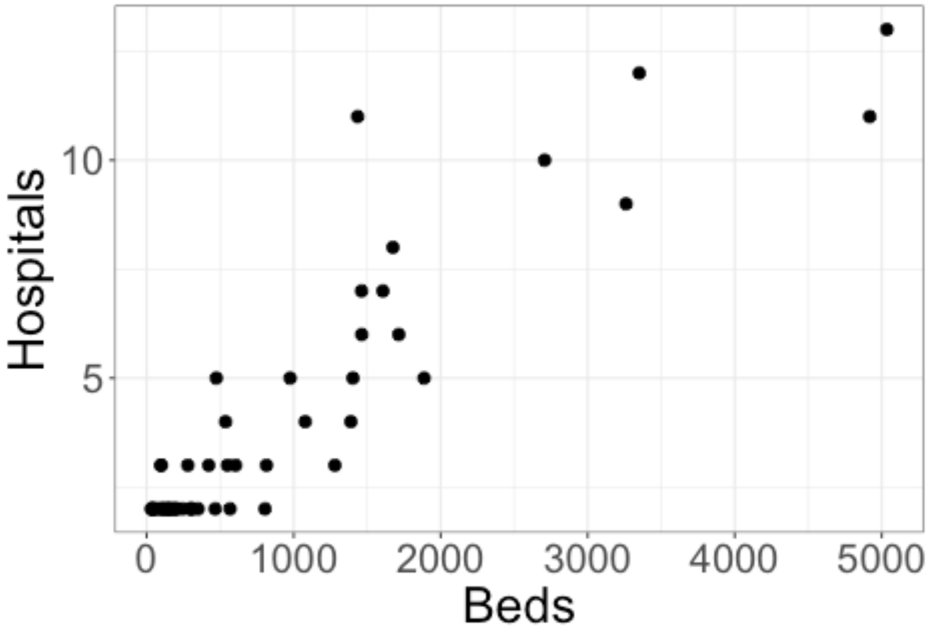
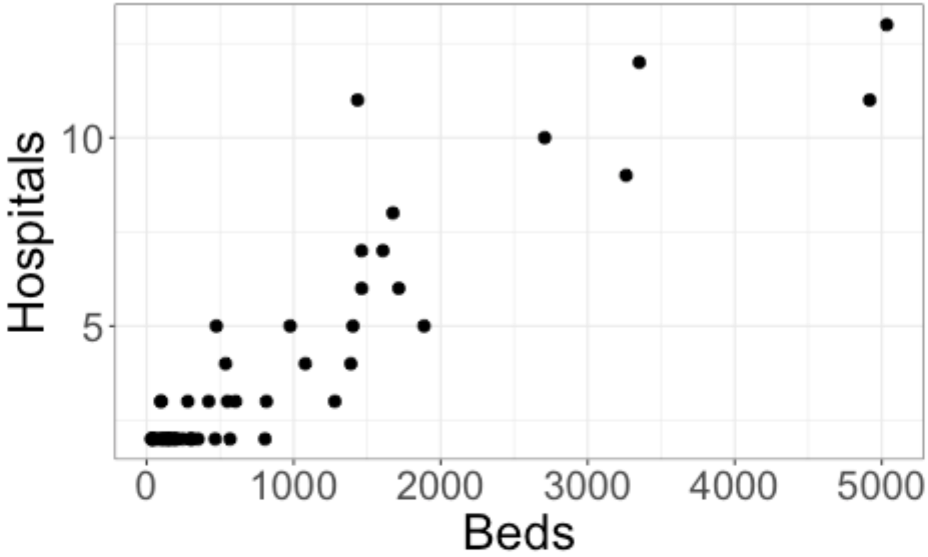


Multicollinearity

Beds and Hospitals are strongly related, so using both doesn't provide much more information than using just one.



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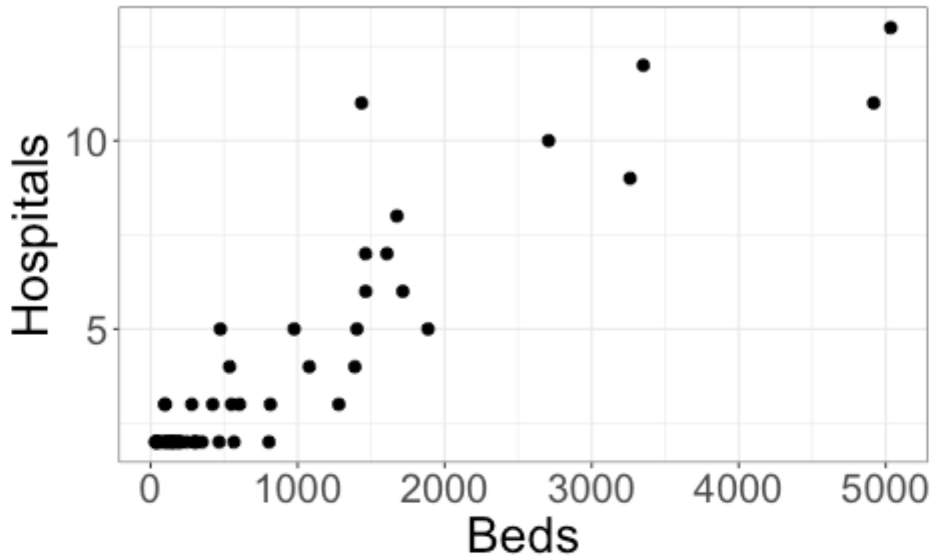
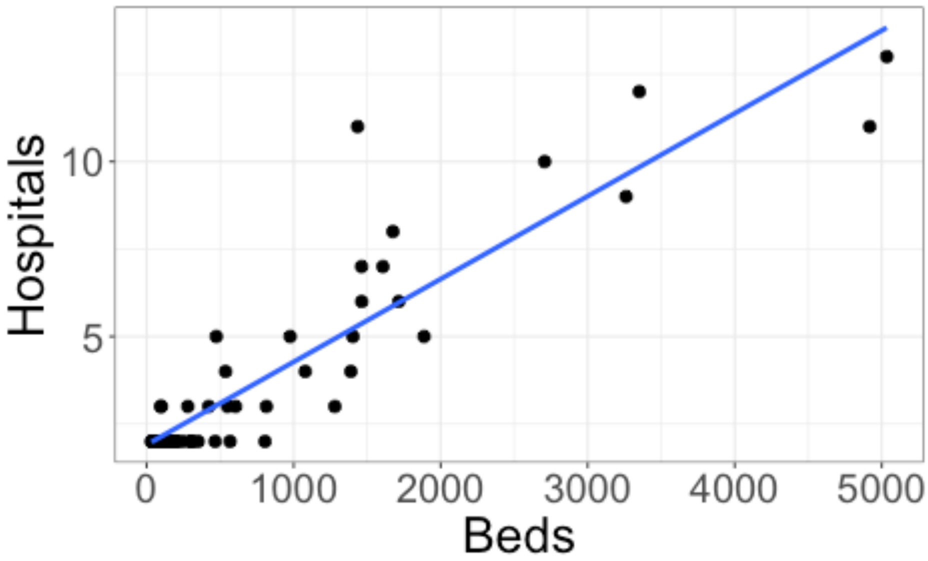


$r = 0.91$ (correlation between Beds and Hospitals)

Measuring multicollinearity



$$\widehat{\text{Beds}} = -514.68 + 349.15 \text{ Hospitals}$$

$R^2 = 0.82$ (for predicting Beds from Hospitals) $= 0.91^2$ $(r = 0.91)$

Variance inflation factor: $VIF = \dfrac{1}{1-R^2} = \dfrac{1}{1-0.82} = 5.56$

Variance inflation factors

Motivation: High multicollinearity means a predictor can be estimated well by some combination of other predictors

Variance inflation factor: Let X_i be a predictor in the model, and R_i^2 the coefficient of determination for a model estimating X_i from the other predictors. The *variance inflation factor* of X_i is

$$VIF_i = \frac{1}{1 - R_i^2}$$

- Higher VIF means more multicollinearity
- Rule of thumb: worried when $VIF_i > 5$ (i.e. $R_i^2 > 0.8$)

Class activity

Work on the class activity. Submit your rendered HTML file at the end of class.

Class activity

- $\widehat{\text{Hgt97}} = 40.59 + 1.10\ \text{Hgt96} \qquad R^2_{adj} =$

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- $\widehat{\text{Hgt97}} = 40.59 + 1.10\,\text{Hgt96} \qquad R^2_{adj} = 0.949$
- $\widehat{\text{Hgt97}} = 40.24 - 1.26\,\text{Diam96} + 1.12\,\text{Hgt96} \qquad R^2_{adj} =$

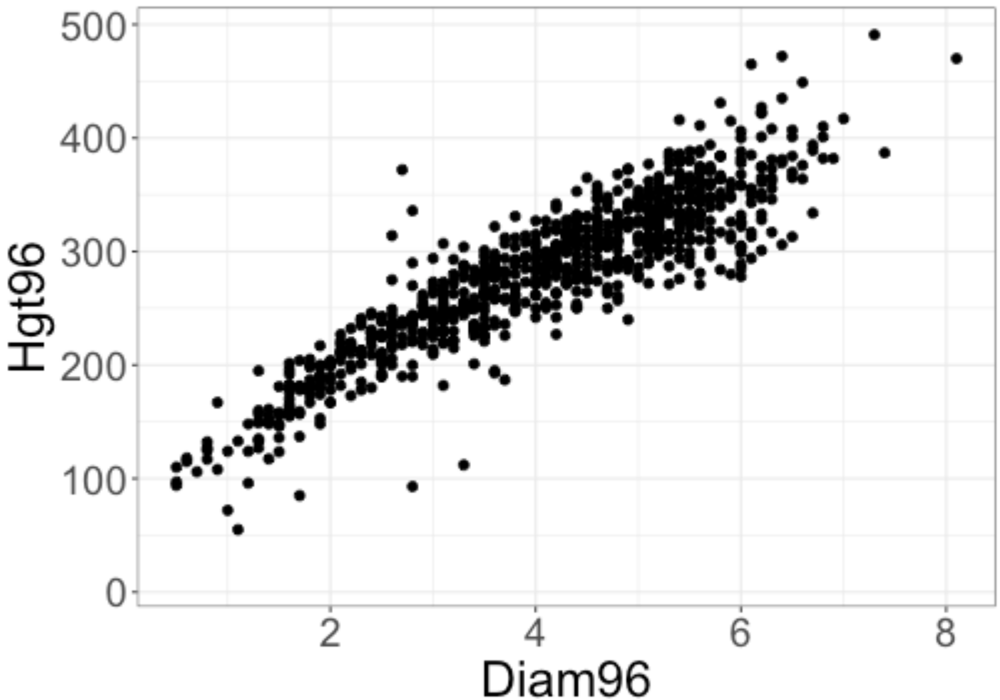
Class activity

- $\widehat{\text{Hgt97}} = 40.59 + 1.10 \text{ Hgt96} \qquad R^2_{adj} = 0.949$
- $$\widehat{\text{Hgt97}} = 40.24 - 1.26 \text{ Diam96} + 1.12 \text{ Hgt96} \qquad R^2_{adj} = 0.942$$

> Why doesn't R^2_{adj} increase when I add diameter to the model?

Class activity

> Why doesn't R^2_{adj} increase when I add diameter to the model?



Because diameter and height in 1996 are so correlated.

Class activity

$$\widehat{\text{Hgt97}} = 40.24 - 1.26\ \text{Diam96} + 1.12\ \text{Hgt96}$$

What is the variance inflation factor for Diam96?

Class activity

$$\widehat{\text{Hgt97}} = 40.24 - 1.26\ \text{Diam96} + 1.12\ \text{Hgt96}$$

What is the variance inflation factor for Diam96?

```
m1 <- lm(Diam96 ~ Hgt96, data = Pines)
summary(m1)$r.squared
```

```
## [1] 0.8134518
```

$$VIF = \frac{1}{1 - 0.813} = 5.35$$

Why is multicollinearity a problem?

- Inflates variability of estimated coefficients
- Interpretation of individual terms is difficult

$$\widehat{\sqrt{\mathrm{MDs}}} = 3.58 + 2.58\ \mathrm{Hospitals} + 0.012\ \mathrm{Beds}$$

Usual interpretation: Holding Beds fixed, an increase of 1 hospital is associated with an increase of 2.58 units in $\sqrt{\mathrm{MDs}}$

Problem: When Beds and Hospitals are highly correlated, doesn't make sense to fix one and change the other. (Can't really change the number of hospitals without changing the number of beds)